

JASKARAN VIRK

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SUMMARY

AI/ML Software Engineer specializing in Generative AI & LLM Application — building RAG pipelines, LLM-integrated apps, and cloud-native backends on a full-stack foundation. Pursuing BCA at Chandigarh University (expected 2027)

SKILLS

AI/ML & GenAI: Machine Learning, Deep Learning, LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, PyTorch, scikit-learn, Ollama, ChromaDB, Gemini

Data & Analysis: NumPy, Pandas, Seaborn, Matplotlib

Programming Languages: Python, JavaScript (ES6+), TypeScript, Java

Databases: PostgreSQL, MongoDB, MySQL, Supabase

Web Development: React.js, Next.js, FastAPI, Node.js, Express.js, REST APIs

Tools & Platforms: Git, GitHub, Postman, VS Code, Vercel

PROJECTS

AskDOCS — RAG-Based Document Q&A System

Tech Stack: FastAPI, PostgreSQL, Gemini, MistralAI, Streamlit | [Live Demo](#) | [GitHub](#)

- Cut irrelevant and hallucinated answers in document Q&A, as measured by grounded, source-matched responses across multi-document tests, by building a 6-stage RAG pipeline with Gemini 3072-D embeddings and semantic retrieval.
- Delivered a secure, multi-user backend built for production traffic, as measured by authenticated access across every endpoint, by architecting a cloud-native FastAPI service with PostgreSQL (pgvector) and Supabase Storage in a modular service/repository pattern.

SnapClass — AI-Powered Smart Attendance System

Tech Stack: React.js, Streamlit, FastAPI, Python, Supabase, dlib, scikit-learn (SVM) | [Live Demo](#) | [GitHub](#)

- Cut manual attendance-taking effort by roughly 95%, as measured by time-to-complete-roster per class session, by replacing manual roll-call with a dlib-embeddings + SVM facial-recognition pipeline reaching 98% recognition accuracy.
- Delivered a fast, reliable attendance backend, as measured by an 80% drop in processing time versus manual methods, by building a FastAPI service integrated with Supabase for real-time image capture and record storage.

Resumind — AI-Powered Resume Analyzer

Tech Stack: React.js, Puter.js | [Live Demo](#) | [GitHub](#)

- Improved resume-feedback accuracy by 40% over baseline manual review, as measured by user-rated relevance of suggestions, by building an AI-based feedback engine with Puter.js and React.js.
- Cut visual-parsing errors by 70% and lifted perceived resume quality by 50% in user testing, by automating resume-to-image conversion and delivering real-time, role-specific optimization suggestions.

Splitr — AI-Powered Expense Management App

Tech Stack: Next.js, Convex, Clerk Auth, ShadCN UI, Cron Jobs | [Live Demo](#) | [GitHub](#)

- Improved app responsiveness and load speed by 40%, as measured by page-load and sync benchmarks, by building a full-stack expense app with Next.js, Convex, and Clerk Auth for instant data sync.
- Cut manual expense-reconciliation effort by 80% and increased user engagement, by implementing transparent expense tracking, instant settlements, real-time dashboards, and scheduled summaries via cron jobs.

EDUCATION

Bachelor of Computer Applications — Chandigarh University | CGPA: 8.62 | Expected 2027

XII (PSEB) — Genius Public School, Sardulgarh | 91.0% | 2023

X (PSEB) — Genius Public School, Sardulgarh | 100.0% | 2021

CERTIFICATIONS & ADDITIONAL

- AI/ML Engineering — Apna College (2026)
- Full-Stack Web Development — Apna College (2025)
- Data Structures & Algorithms — Apna College (2025)
- Solved 140+ algorithmic problems on LeetCode